Jaehyun Yoon

Mechanical Engineering | Robotics Researcher

Korea University | yjh54280@korea.ac.kr | GitHub | Portfolio

Research Interests

Continuum Robotics • Medical Robotics • Bio-Inspired Robotics • Compliant Mechanisms • Smart Material Actuation • Haptic Interfaces

Education

Korea University **2026 – Present**
M.S. in Mechanical Engineering *Seoul, Republic of Korea*

Konkuk University **2019 – 2026**
B.S. in Mechanical Engineering *Seoul, Republic of Korea*

Research Experience

Korea University **2026 – Present**
Graduate Researcher, Department of Mechanical Engineering *Seoul, Republic of Korea*
Advisor: Dahyun (Daniel) Lim

- Developing **prehensile tail-inspired locking joint modules** with SMA-actuated unlocking and tendon-driven actuation for selective shape reconfiguration and power-free posture maintenance.

Konkuk University **Jan. 2025 – Dec. 2025**
Undergraduate Researcher, Department of Mechanical Engineering *Seoul, Republic of Korea*
Advisor: Tae-Heon Yang

- Developed a **portable commerce photography automation robot** for consistent multi-view product imaging.
- Contributed to **MORPH**, an MR-fluid-based reconfigurable haptic grip interface for bidirectional and multimodal grasping interaction in telerobotics and XR.

Publications

- **J. H. Yoon**, J. S. Bae, J. H. Choi, S. Kang, J. Lee, and T. H. Yang, "Development of a Portable Commerce Photography Automation Robot," *Journal of the Korean Society for Precision Engineering*, vol. 43, no. 7, pp. 689–700, 2026. [DOI]
- "MORPH: MR-Fluid Operated Reconfigurable Programmable Haptic Grip Interface for Bidirectional and Multimodal Grasping Interaction in Telerobotics and XR," *Advanced Intelligent Systems*, **Under Review**.
- "A Parametrically Scalable Compliant Gripper for Fruit Harvesting: Mold-Free Additive Manufacturing with Reduced Material and Process Waste," *International Journal of Precision Engineering and Manufacturing-Green Technology*, **Under Review**.

Conference Presentations

- "Prehensile Tail-Inspired Locking Joint Modules for Power-Free Shape Retention," *2026 Korean Society of Mechanical Engineers (KSME) Fall Conference*, Nov. 2026. **Scheduled oral presentation**.

Awards & Honors

Runner-Up (2nd Place), 2026 Korea University K-3MT Competition **2026**
2nd place among 45 graduate student participants *Korea University*

- Selected as Runner-Up for effectively communicating graduate research to a non-specialist audience in a three-minute presentation.

Junior Fellowship Research Grant (JFR) **2026**
Research Grant Recipient, Graduate School *Korea University*

Technical Skills

CAD & Mechanical Design	Fusion 360, Autodesk Inventor, CATIA
Rapid Prototyping	FDM, SLA, Additive Manufacturing, Mechanical Prototyping
Robotics	Continuum Robots, Tendon-Driven Mechanisms, Compliant Mechanisms, Mechanical Locking Mechanisms
Smart Materials	Shape Memory Alloy (SMA), Magnetorheological (MR) Fluid
Experimental Systems	Laser Displacement Sensing, Thermal Imaging, Current/Voltage Measurement, Robotic Experimentation
Research Areas	Medical Robotics, Bio-Inspired Robotics, Haptics, Smart-Material Actuation